Do More Retrieved Web Documents Improve RAG? Retrieval Saturation on the Open Web Search Curlie 2025 Collection

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Abstract

Retrieval-augmented generation grounds language-model answers in retrieved documents, which reduces hallucination. It is often assumed that retrieving more documents yields better answers—but how much do we actually need to retrieve? We study when generation quality saturates on noisy open-web data, and whether this behaviour differs between sparse and dense retrieval. We observe a descriptive plateau in generation quality at approximately three retrieved documents for both retrievers. Beyond that point, retrieving more neither improves nor degrades the answers, and in our pooled exploratory analysis, retrieval noise has near-zero rank correlation with generation quality. This near-zero correlation indicates no detectable monotonic association in our data, but it cannot rule out an effect of retrieval noise.

1 Introduction

Retrieval-augmented generation (RAG) improves the answers of a large language model (LLM) by retrieving external documents and adding them to the model's context, enabling the model to produce grounded answers and reducing hallucination. A common assumption is that retrieving more documents leads to better answers, since the relevant passage is then more likely to be somewhere in the context. In practice, however, more documents also mean more noise and more compute, so it is not obvious that they improve answer quality.

Recent work suggests they do not always. On curated corpora, generation quality can first improve and then decline as more passages are added, because the extra passages include hard negatives that look relevant but do not support the answer [2, 7]. Most of these studies use clean collections such as Wikipedia. It is less clear whether the same saturation appears on noisy, web-derived data, where retrieved documents are messier and less reliably relevant. We study this on the `radboud-val` split of the Open Web Search Curlie 2025 collection [10], an openly available web corpus that lets us build and account for every step of the pipeline ourselves rather than treating retrieval as a black box.

Concretely, we retrieve the top documents for each topic with a sparse (BM25) and a dense retriever, feed increasing numbers of them (k) to a locally hosted Gemma generator, and evaluate the answers with RAGAS faithfulness and answer relevancy, complemented by a manual audit and an analysis of retrieval noise. Our main hypothesis is:

> *Increasing the number of retrieved documents improves retrieval recall, but does not necessarily improve generated answer quality. After a certain retrieval depth, additional documents may introduce noise or hard negatives that reduce answer quality.*

This leads to three research questions:

RQ1 Does retrieval saturation occur on the `radboud-val` corpus? We test whether generation quality improves, plateaus, or decreases as k increases.

RQ2 Does saturation behaviour differ between sparse and dense retrieval?

RQ3 Is lower generation quality associated with noisier retrieved contexts?

2 Related Work

Retrieval depth and hard negatives. Jin et al. [7] study how the number of retrieved passages affects long-context RAG and find an inverted-U: quality first improves and then declines as more passages are added, which they attribute to hard negatives—documents that appear relevant but do not support the answer. The effect is strongest with a strong dense retriever (e.g. E5), which ranks such hard negatives highly, whereas a weaker lexical retriever (BM25) tends to plateau. Their experiments use curated data, which motivates our question of whether the same behaviour appears on noisy web documents.

More context is not always better. Liu et al. [8] show that giving a language model longer input is a trade-off: more information can help, but it also increases how much the model must reason over and can reduce accuracy. Their study uses Wikipedia and open-domain QA, which are curated and do not reflect real web data—exactly the gap our work targets.

The role of noise. Cuconasu et al. [2] show that the documents surrounding the relevant one matter: in their setting, adding random documents to the context improved accuracy by up to 35%, yet they also find that retrieving only

about three to five documents is most effective, since beyond that the risk of including distracting documents grows. Because our experiment adds more documents—and thus more noise—as k increases, this directly motivates our analysis of whether retrieval noise is associated with lower answer quality.

3 Method

We study how the number of retrieved documents affects the quality of the answers a RAG system generates. We run this experiment over a set of *topics*; each topic is a search query, together with human relevance judgements marking which documents are relevant to it. For each topic we retrieve the top 50 documents with two retrievers—a sparse one (BM25) and a dense one—and feed the generator increasing numbers of them (the top k, for growing k), to see whether answer quality keeps improving or saturates at some point (RQ1). Both retrievers run under the same conditions—the same topics, prompt, generator, and values of k—so that any difference in saturation can be attributed to the retrieval method and not to the setup (RQ2). For every combination of topic, retriever, and k we generate the answer with the same prompt and decoding settings, so the answers stay comparable. The prompt tells the model to use only the retrieved documents and to refuse when they do not contain the answer, which keeps the answers grounded and limits hallucination. We then evaluate each answer on two aspects: *faithfulness*, whether it is grounded in the retrieved documents, and *answer relevancy*, whether it actually addresses the query. Because these scores are themselves produced by an LLM, we validate them against a small blind manual audit: on a stratified sample we assign our own labels without seeing the automatic scores, then check that the two agree. Finally, since adding more documents also adds more possibly non-relevant or unjudged content, we measure the retrieval noise at each k and test whether it is associated with lower answer quality [2, 7] (RQ3).

We deviate from our original plan in one respect: instead of the precomputed Jina V3 document embeddings shipped with the split, we embed both the documents and the queries ourselves with a pinned open model, because the recipe needed to encode the queries consistently with those embeddings was not available and mixing the two would introduce an uncontrolled confound. The exact model and settings are given in Section 4.

4 Experiments

Data. We use the `radboud-val` validation split of the Open Web Search Curlie 2025 collection [10], accessed through the `ir-datasets-owi` package [9]. The split provides the topics (queries), the document collection, and human relevance judgements (qrels); our evaluation uses the 28 judged topics.

Retrieval. For sparse retrieval we use BM25 [11] over the document text.For dense retrieval we embed the documents ourselves with a pinned open embedding model (Jina v5-nano) [1],[1] which produces 768-dimensional, L2-normalised embeddings. Documents are split into 1024-token chunks with 128-token overlap before encoding, and queries and documents are encoded asymmetrically. We index the chunk embeddings with an exact FAISS inner-product index [3]; because the vectors are unit-length, inner product equals cosine similarity, and each document is scored by its best-matching chunk. For every topic we cache the top 50 retrieved documents from each retriever. Retrieval quality is measured with Recall@k and nDCG@k [6], computed with the ir-measures package, against the qrels.

Generation. We generate answers with the instruction-tuned Gemma 4 model `gemma4:e4b` [5], served locally through Ollama. Because it is a reasoning model, we disable its hidden reasoning (`think` = false) so it answers directly. Decoding is greedy (temperature 0) for reproducibility, with a context window of 8192 tokens and up to 512 generated tokens. Each retrieved document is truncated to its first 500 characters before being placed in the context, so that even the deepest setting ($k = 50$) fits the context window. The prompt is fixed across all runs:

> *You are writing a brief answer to a search query. Use ONLY information from the documents below; do not use outside knowledge. If the documents do not contain the answer, reply exactly: "The documents do not answer this." Refer to documents inline as [Document N]. Keep the answer to 2–4 sentences.*

followed by the query, the labelled `[Document N]` context blocks, and an `Answer:` cue. We run this for every combination of topic, retriever, and $k \in \{1, 3, 5, 10, 20, 50\}$, producing $28 \times 2 \times 6 = 336$ cached answers.[2]

Evaluation. We score every answer with RAGAS [4] Faithfulness and Answer Relevancy. The judge is the same Gemma 4 model, but with a larger 16384-token context window so that the longest prompts are not truncated; the embedder used by Answer Relevancy is `nomic-embed-text`, also served through Ollama. Because the prompt lets the model refuse when the documents do not answer the query, faithfulness is undefined for refusals and we record it as NaN. To check that the automatic scores are trustworthy, we additionally run a small blind manual audit on a stratified sample of answers. Finally, for RQ3 we quantify retrieval noise from the qrels. Because a share of the retrieved documents are

[1] Exact model: `jinaai/jina-embeddings-v5-text-nano-retrieval`, Hugging Face revision `ac5d898c8d382b17167c33e5c8af644a3519b47d`.
[2] Code, prompts, ranking files, cached generations, and evaluation artefacts are available at https://git.fim.uni-passau.de/nemeth/data-science-seminar/-/tree/37d98306c29ec93a66027620bbb1c57eae7e1e17.

unjudged—and this share grows with k—we report both an inclusive and a strict version, as an upper and a lower bound:

$$\text{noise} = \frac{\text{non-relevant} + \text{unjudged}}{k}, \tag{1}$$

$$\text{noise}_{\text{strict}} = \frac{\text{non-relevant}}{k}. \tag{2}$$

5 Results

We first look at retrieval quality (Table 1). As expected, Recall@k increases with k, but only up to about 0.65 for sparse and 0.62 for dense at $k = 50$, and nDCG@k stays roughly between 0.38 and 0.57. Sparse retrieval (BM25) outperforms dense at every value of k, on both Recall@k and nDCG@k.

Retrieval noise grows steadily with k (Figure 1), but almost all of the growth is in *unjudged* documents rather than confirmed non-relevant ones: the strict noise stays low and roughly flat, while the unjudged share rises sharply. For BM25, strict noise is 0.25 at $k = 1$ and only 0.13 at $k = 50$, whereas the unjudged share grows to 0.68, giving an inclusive noise of 0.81 at $k = 50$.

Answer quality, in contrast, does not track the noise (Figure 2). Faithfulness is high and flat across all k, with RAGAS scores between 0.92 and 1.00. Answer relevancy is lowest at $k = 1$ (0.61 for sparse, 0.67 for dense), rises to about 0.78 at $k = 3$, and then plateaus for all larger k. The pooled row-level Spearman correlation between our qrel-based noise *proxy* and quality is effectively zero (faithfulness $\rho = +0.01$, $n = 322$; answer relevancy $\rho = +0.02$, $n = 336$). We read this as exploratory: within our setup we find no detectable association between noise and judged quality, though this does not establish that noise has no effect (RQ3).

To check whether the RAGAS scores can be trusted, we manually rated a sample of 24 answers (2 per retriever $\times k$ cell) without looking at the RAGAS scores, and then compared the two. For faithfulness the agreement is strong: we rated all 24 answers as "Yes", and RAGAS scored them 0.98 on average (minimum 0.80). For answer relevancy the agreement is moderate: RAGAS averaged 0.82 on the answers we rated "Yes" versus 0.69 on those we rated "Partly" (Spearman +0.48), i.e. RAGAS is slightly lenient on borderline cases.

6 Analysis

Faithfulness is high and flat across all k because it measures whether the answer is grounded in the retrieved context, not whether it is factually correct; a well-grounded answer built from a non-relevant document still scores as faithful. Answer relevancy tells the more interesting story: it reaches its best value at around three documents and then plateaus (RQ1). The top-ranked documents already contain what is needed to answer the query, so the additional documents mostly add noise rather than new useful content, and the answers do not improve. This is consistent with the near-zero correlation between noise and quality (RQ3): within our setup the extra, noisier context neither clearly helps nor clearly hurts.

Table 1. Retrieval quality for sparse and dense retrieval.

	Recall@k		nDCG@k	
k	BM25	Dense	BM25	Dense
1	**0.062**	0.048	**0.494**	0.375
3	**0.152**	0.147	**0.489**	0.454
5	**0.233**	0.210	**0.496**	0.438
10	**0.386**	0.314	**0.502**	0.411
20	**0.504**	0.465	**0.513**	0.461
50	**0.651**	0.623	**0.566**	0.520

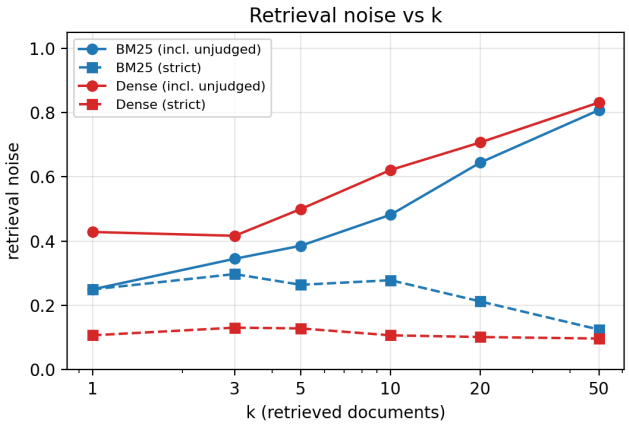


Figure 1. Retrieval noise vs. k: inclusive (non-relevant + unjudged) and strict (confirmed non-relevant only).

The saturation we observe looks like a plateau rather than the inverted-U reported for stronger retrievers [7]. A likely reason is that both of our retrievers are mediocre—nDCG peaks at only about 0.57 (RQ2)—so they rarely surface the kind of convincing hard negatives that actively mislead the generator; the extra documents are more often simply unhelpful than harmful. Moreover, most of the growth in noise is in unjudged rather than confirmed non-relevant documents, so this robustness observation is limited to that kind of noise—a point we return to in the limitations.

7 Limitations

Our study has several concrete limitations. First, to fit even the deepest setting ($k = 50$) into the generator's context window, we truncate every retrieved document to its first 500 characters. This removes most of the text of longer documents, so our finding that additional documents do not help may partly reflect that the generator only ever sees short prefixes; whether full-document context changes the picture is untested.

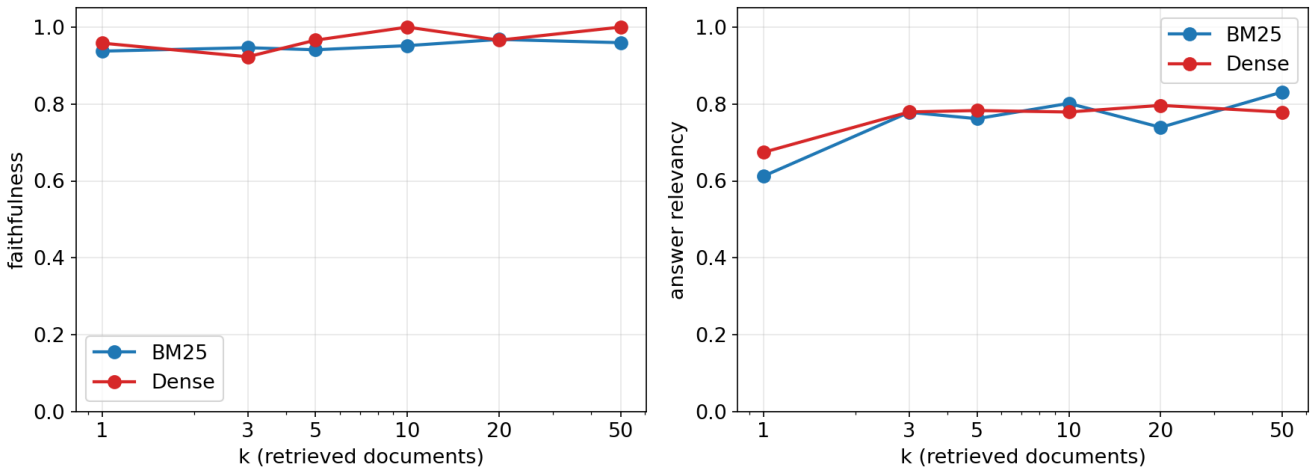


Figure 2. Faithfulness and answer relevancy vs. k for BM25 and dense retrieval (dense = Jina v5-nano embeddings + FAISS).

Second, both of our retrievers are weak—nDCG peaks at about 0.57. We therefore cannot observe the inverted-U degradation reported for strong retrievers [7], because our retrievers rarely surface convincing hard negatives. Our conclusion is thus that quality *plateaus* on this corpus with these retrievers, not that RAG is generally robust to noisier context.

Third, the noise we add at high k is dominated by *unjudged* documents rather than confirmed non-relevant ones, because the qrels pool only shallow ranks. The near-zero noise–quality correlation therefore shows robustness to unjudged documents whose relevance is unknown, not to adversarial hard negatives.

Fourth, the evaluation itself is limited: the automatic metrics are reference-free, so faithfulness measures grounding rather than factual correctness—a confident but wrong answer built from a retrieved document can still score as faithful. The manual audit that calibrates these scores covers only 24 answers rated by a single annotator, so it indicates plausibility but is not a strong validation.

Finally, we evaluate only the `radboud-val` split (28 judged topics), a small validation subset of the collection; the results are evidence for retrieval saturation on this manageable web-derived corpus, not a claim about the full Open Web Search collection or the web in general.

8 Conclusion

We asked whether retrieval saturation appears on noisy open-web data and whether it differs between sparse and dense retrieval. On the `radboud-val` split, we observe that generation quality saturates early: answer relevancy is best at around three documents and then plateaus, faithfulness stays high and flat, and our qrel-based noise proxy is not associated with lower judged quality on this split. Saturation appears for both BM25 and dense retrieval, with BM25 the stronger retriever throughout. In short, beyond a small number of documents, retrieving more does not improve the answers on this corpus.

For future work, the most direct next step is to repeat the experiment with a stronger retriever, to test whether the plateau turns into the inverted-U degradation seen on cleaner corpora [7]. With more compute, one could also feed whole documents instead of 500-character snippets, use a larger generator model, and categorise the retrieved noise—for example separating hard negatives from merely unjudged documents—to see what drives the small dip we observe around $k = 20$.

Disclaimer on the Use of AI Tools

I used a generative AI assistant in this project. For this report, it helped with LATEX and ACM formatting, polishing the writing, and generating and relabeling the figures. It was also used during development of the code, for writing and refactoring pipeline code, documentation, and debugging. The research design, experiments, evaluation, and interpretation of the results are my own.

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